

CRIME INCIDENT DATA ANALYSIS AND DASHBOARD (2020–2024)

1. Project Overview

This mini project focuses on analyzing crime incidents and crime patterns using comprehensive crime datasets and Power BI dashboards. The analysis monitors crime occurrences across different areas, victim demographics, weapon usage, crime status, and hotspot locations using various visualization techniques.

Using Microsoft Excel and Power BI, the project converts raw crime records into meaningful insights through systematic data cleaning, data transformation, KPI analysis, and interactive dashboard reporting. The primary goal is to provide meaningful crime insights that support data-driven decision-making and help identify crime trends and hotspot locations through visualization and analytics.

2. Objectives

- Analyze crime distribution across different areas
- Monitor victim demographics and gender patterns
- Track crime type occurrences and frequency
- Analyze weapon usage trends in crime incidents
- Identify crime hotspot locations using maps
- Compare crime occurrence patterns across categories
- Build interactive Power BI dashboards
- Generate actionable insights through data analysis

3. Data Sources

Dataset Information

- Dataset Source: Government Crime Dataset
- Domain: Crime Analytics
- Timeline: 2024 – 2026
- Data Size: 10,000+ records and 20+ columns

Datasets Fields Used:

- DR_NO
- DATE OCC
- AREA NAME
- Vict Age
- Vict Sex
- Weapon Desc
- Status Desc
- Premis Desc
- LAT
- LON

4. Data Cleaning & Transformation

Data Cleaning Steps

- Removed duplicate records
- Eliminated unnecessary columns
- Corrected inconsistent data types
- Renamed columns for improved readability
- Handled missing and null values
- Processed latitude and longitude values
- Removed invalid records

Data Transformation Steps

- Created KPI measures
- Applied count aggregation using DR_NO
- Developed slicers and filters
- Built hotspot mapping using geographical values
- Created interactive visual dashboards

KPI Measures Created

- Total Crimes
- Crime Frequency

- Victim Count
- Weapon Usage Count
- Crime Hotspots
- Area-wise Crime Count

Metrics Created

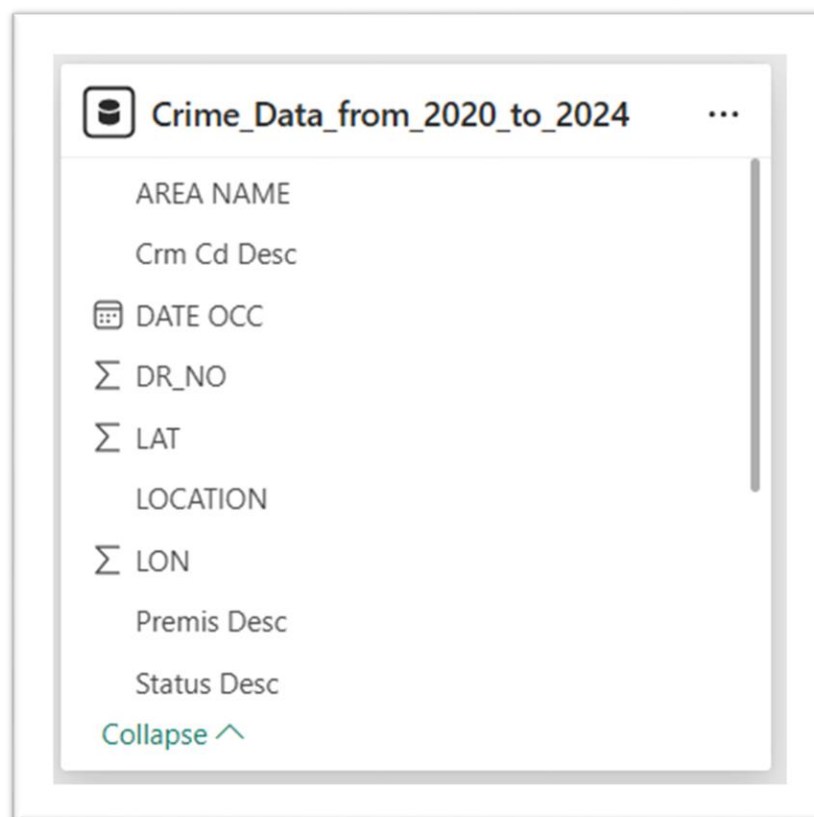
- Total Crimes KPI
- Total Areas KPI
- Victim Distribution %
- Crime Frequency %
- Weapon Usage %

Tools & Technologies Used

- Microsoft Power BI
- Microsoft Excel
- Power Query
- Data Visualization Techniques

Data Modeling

- A single-table data model was used for this project.
- Since the project used one primary crime dataset, complex table relationships were not required.
- The dataset fields were directly connected with Power BI visualizations and KPI measures.
- Filters and slicers were applied to create interactive dashboard analysis.
- The data model structure supported efficient dashboard creation and report generation.

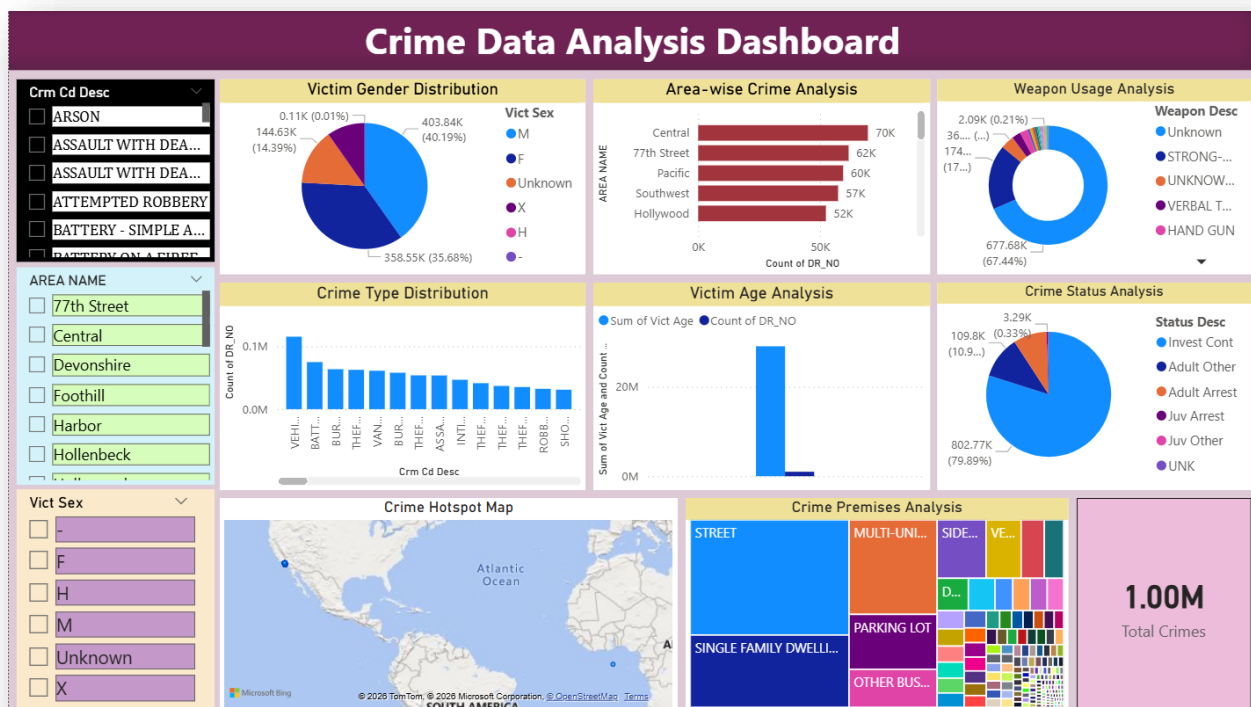


5. Dashboard Overview

The dashboard was designed using Power BI to provide interactive and visual analysis of crime data. Different charts and visualizations were used to analyze crime trends, victim information, crime locations, and hotspot areas.

Dashboard Features:

- KPI Cards
- Slicers and Filters
- Interactive Charts
- Crime Hotspot Maps
- Visual Trend Analysis



6. Key Insights & Findings

Objective 1: Area-wise Crime Analysis

Descriptive Analytics

Certain areas recorded a higher number of crime incidents compared to other areas.

Diagnostic Analytics

Population density and environmental factors may contribute to increased crime activity.

Predictive Analytics

High-crime areas may continue showing increased crime activity if trends remain unchanged.

Prescriptive Analytics

Improved monitoring and safety measures should be implemented in high-crime areas.

Objective 2: Victim Gender Distribution

Descriptive Analytics

Crime incidents were distributed across male and female victims.

Diagnostic Analytics

Demographic and social factors may influence victim distribution patterns.

Predictive Analytics

Victim distribution trends are expected to remain similar in future periods.

Prescriptive Analytics

Safety awareness and targeted support programs can be improved.

Objective 3: Crime Type Distribution

Descriptive Analytics

Certain crime categories occurred more frequently than others.

Diagnostic Analytics

Environmental and location-based factors may influence crime types.

Predictive Analytics

Frequently occurring crime categories may continue increasing.

Prescriptive Analytics

Focused prevention strategies can help reduce repeated crime categories.

Objective 4: Weapon Usage Analysis

Descriptive Analytics

Some weapons were used more frequently in reported incidents.

Diagnostic Analytics

Weapon usage may vary depending on crime category and location.

Prescriptive Analytics

Improved security measures and preventive actions can help reduce weapon-related incidents.

Objective 5: Crime Hotspot Analysis

Descriptive Analytics

Hotspot locations showed higher concentrations of crime incidents.

Diagnostic Analytics

Location-based patterns indicate higher crime occurrence in specific areas.

Predictive Analytics

Hotspot regions may continue experiencing high crime activity.

Prescriptive Analytics

Surveillance systems and law enforcement monitoring can be strengthened in hotspot locations.

7. Conclusion

This mini project successfully analyzed crime patterns and crime-related trends using Power BI dashboards and structured crime datasets. The analysis identified area-wise crime differences, victim demographics, weapon usage trends, and crime hotspot locations.

Interactive visualizations helped transform raw crime data into meaningful insights and demonstrated how Power BI supports effective data-driven analysis and decision-making.